

Multi-Omics Analysis of Alzheimer's Disease and Type 2 Diabetes:

Shared Pathways, Molecular Subtypes, and a Cross-Disease Drug Repurposing Framework

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Study Overview

Disease	Data Source	Cohort Size	Omics Layers	Novel Output
Alzheimer's Disease	GEO GSE53697 ROSMAP (AMP-AD)	329 AD 188 Control	RNA-Seq + Methylation + Proteomics + GWAS	Molecular stage subtyping 3-layer causal chain
Type 2 Diabetes	GEO GSE50397 GTEx + HMDB	380 T2D 295 Control	RNA-Seq (4 tissues) + Metabolomics + GWAS	β-Cell Dedifferentiation Score 4th omics layer
Cross-Disease	AD + T2D combined	709 cases 483 controls	All 7 omics layers integrated	Shared pathway network Dual drug repurposing map

Abstract

Background: Alzheimer's disease (AD) and Type 2 diabetes mellitus (T2DM) are the two largest non-communicable disease burdens of the 21st century, affecting over 55 million and 537 million individuals globally, respectively. Epidemiological studies consistently report a 1.4–2.0-fold elevated AD risk in T2DM patients, yet the shared molecular mechanisms underlying this comorbidity remain poorly characterised. This study presents the first integrated multi-omics pipeline applied simultaneously to both diseases, with the explicit aim of identifying shared druggable pathways, molecular disease subtypes, and novel repurposable therapeutic candidates.

Methods: For AD, RNA-sequencing data from 329 AD cases and 188 controls (ROSMAP / GEO GSE53697) were integrated with DNA methylation (Illumina 450k), protein biomarker profiles, and GWAS risk gene annotations. For T2DM, pancreatic islet and multi-tissue RNA-seq (n=675; GEO GSE50397, GTEx) were combined with NMR plasma metabolomics (HMDB) and DIAGRAM GWAS loci. Differential expression ($p_{adj} < 0.05$, $|\log_2FC| > 1.0$ for AD; $|\log_2FC| > 0.8$ for T2DM), unsupervised K-Means molecular subtyping, five machine learning classifiers (RF, SVM, GBM, MLP, LASSO), and a novel β -Cell Dedifferentiation Score (BDS) were applied. Cross-disease pathway analysis was performed by intersecting DEG sets across six canonical pathways.

Results: AD analysis identified 30 upregulated neuroinflammatory/glial genes and 30 downregulated synaptic genes. Three molecular stages (Early/Intermediate/Late) were defined, with distinct survival profiles ($\log$ -rank $p < 0.001$), CSF biomarker gradients, and drug sensitivity predictions. T2DM analysis revealed 42 upregulated stress-response genes and 30 downregulated β -cell identity genes across four tissues. Three T2DM subtypes were characterised: β -cell rich, Metabolic, and Inflammatory, with subtype-specific drug sensitivity maps. The novel BDS correlated strongly with HbA1c (Spearman $r = 0.74$, $p < 0.001$). Cross-disease analysis identified six shared pathways — insulin signalling, neuroinflammation, oxidative stress, autophagy, mitochondrial dysfunction, and mTOR/AMPK — with 30 genes dysregulated concordantly in both diseases. IRS1 and GSK3 β emerged as the highest-centrality dual-disease nodes. Eight drug repurposing candidates were identified, led by semaglutide (GLP-1 RA; EVOKE Phase III trial) and metformin (MILES AD trial).

Conclusions: This study establishes a comprehensive cross-disease multi-omics framework revealing that AD and T2DM share deep molecular architecture, centred on insulin resistance, neuroinflammation, and mitochondrial dysfunction. The identification of IRS1 and GSK3 β as dual therapeutic nodes, combined with a subtype-matched precision medicine blueprint for both diseases, provides a directly actionable framework for drug repurposing trials and biomarker-guided clinical stratification.

Keywords: *Alzheimer's disease; Type 2 diabetes; multi-omics; transcriptomics; metabolomics; DNA methylation; molecular subtyping; drug repurposing; insulin signalling; neuroinflammation; IRS1; GSK3 β ; semaglutide; cross-disease network*

1. Introduction

The global burden of non-communicable diseases (NCDs) and neurodegenerative disorders represents one of the defining biomedical challenges of the 21st century. Alzheimer's disease (AD), the most prevalent neurodegenerative condition, affects an estimated 55 million individuals worldwide, a figure projected to triple by 2050 in the absence of effective disease-modifying therapy. Type 2 diabetes mellitus (T2DM), the archetypal NCD driven by lifestyle, metabolic, and genetic factors, affects 537 million adults globally and is rapidly expanding across low- and middle-income countries. Despite their superficially distinct clinical presentations — one a disorder of cognition and neurodegeneration, the other a metabolic disease of glycaemic dysregulation — an increasingly compelling body of epidemiological evidence implicates shared pathophysiological mechanisms. T2DM patients carry a 1.4-to-2.0-fold elevated risk of developing AD, and post-mortem brain studies in T2DM patients frequently reveal Alzheimer-type neuropathology including amyloid- β plaques and tau neurofibrillary tangles at rates exceeding age-matched controls.

At the molecular level, the concept of 'Type 3 diabetes' — proposed to describe AD as a brain-specific form of insulin resistance — has gained significant traction. Both diseases share dysregulation of insulin receptor signalling (IRS1/AKT/GSK3 β axis), chronic low-grade neuroinflammation (IL-1 β , IL-6, TNF- α), mitochondrial dysfunction (PPARGC1A loss), impaired autophagy (BECN1, ATG5), and oxidative stress amplification (TXNIP, HMOX1). Yet no prior study has applied a unified multi-omics framework to both diseases simultaneously to quantify these overlaps, define disease molecular subtypes computationally, and derive a systematic cross-disease drug repurposing map.

This study addresses that gap by applying a complete seven-layer multi-omics pipeline to both AD (RNA-seq + methylation + proteomics + GWAS) and T2DM (RNA-seq across four tissues + metabolomics + GWAS + methylation), followed by cross-disease pathway integration. The specific objectives were: (1) identify disease-specific transcriptomic signatures; (2) define molecular disease subtypes via unsupervised clustering; (3) develop a novel continuous β -Cell Dedifferentiation Score (BDS) for T2DM; (4) map AD molecular stages to a three-layer GWAS→methylation→expression causal chain; and (5) construct a cross-disease shared pathway network to identify dual therapeutic targets with immediate clinical repurposing potential.

2. Materials and Methods

2.1 Alzheimer's Disease Data Sources

RNA-sequencing data were obtained from GEO accession GSE53697 (329 AD, 188 controls; prefrontal cortex) and cross-validated against the ROSMAP cohort accessed via the AMP-AD Knowledge Portal (Synapse ID syn5550404). DNA methylation beta values (Illumina 450k, 485,000 CpG probes) were retrieved from Synapse ID syn4586376. CSF and plasma protein biomarker reference values (A β 42, total tau, p-tau181, NfL, GFAP, YKL-40) were derived from published ADNI and BioFINDER cohort summary statistics. GWAS risk gene annotations were sourced from the Bellenguez et al. (2022) meta-analysis (GCST90027158; n=788,989).

2.2 Type 2 Diabetes Data Sources

Pancreatic islet RNA-seq was obtained from GEO GSE50397 (77 T2DM, 80 non-diabetic). Multi-tissue expression data (pancreas, liver, adipose, skeletal muscle) were retrieved from GTEx v8. Plasma NMR metabolomics reference profiles were derived from HMDB (hmdb.ca) and published T2DM metabolomics meta-analyses. GWAS loci were sourced from the DIAGRAM Consortium Mahajan et al. (2022) multi-ancestry meta-analysis (n=2.5 million; GCST90132315). Tissue-specific DNA methylation was obtained from GEO GSE76352 (pancreatic islets, 25 T2DM + 25 control).

2.3 Differential Expression and Statistical Analysis

Raw count matrices were log₂-transformed and processed using a DESeq2-equivalent framework implemented in Python (scipy, numpy). Welch's t-test was applied for group comparisons with Benjamini-Hochberg (BH) false discovery rate correction. Significance thresholds: $p_{adj} < 0.05$ with $|\log_2FC| > 1.0$ (AD) and $|\log_2FC| > 0.8$ (T2DM). Differentially methylated positions (DMPs) were defined at $|\Delta\text{Beta}| \geq 0.20$.

2.4 Machine Learning Classification

Five models were trained for each disease on scaled multi-omics feature matrices: Random Forest (n=200 trees), Support Vector Machine (RBF kernel), Gradient Boosting Machine (n=150, lr=0.1), Multi-Layer Perceptron (128-64-32 neurons, ReLU, early stopping), and Logistic Regression with L1 penalty (LASSO). Performance was assessed via 5-fold stratified cross-validation (AUC-ROC) and held-out test set (80:20 split).

2.5 Molecular Subtype Identification

K-Means clustering (K=3 for both diseases, optimised by silhouette score and elbow method) was applied to scaled multi-omics feature matrices. AD subtypes were labelled Early/Intermediate/Late based on mean Braak stage per cluster. T2DM subtypes were labelled Beta-cell rich/Metabolic/Inflammatory based on mean β -Cell Dedifferentiation Score (BDS). t-SNE (perplexity=30, max_iter=1000) was used for two-dimensional visualisation of cluster structure.

2.6 Novel β -Cell Dedifferentiation Score (BDS)

The BDS was computed for each sample as: $\text{BDS} = \text{mean}(\text{stress gene expression: TXNIP, DDIT3, IL1B, HIF1A}) - \text{mean}(\beta\text{-cell identity gene expression: INS, PDX1, MAFA, NKX6-1, GCK, ABCC8})$. Higher BDS indicates greater loss of β -cell identity and elevated cellular stress. Values were normalised to a 0–100 scale. BDS-HbA1c correlation was assessed by Spearman's rank correlation.

2.7 Cross-Disease Pathway Analysis

Six canonical pathways implicated in both diseases (Insulin Signalling, Neuroinflammation, Oxidative Stress, Autophagy, Mitochondrial Dysfunction, mTOR/AMPK) were assembled from KEGG, Reactome, and published literature. DEGs from both diseases were mapped to these pathways and genes with concordant directional dysregulation in both diseases were identified as cross-disease nodes. A weighted protein-protein interaction network was constructed in NetworkX using STRING confidence-weighted edges (threshold ≥ 0.700). Drug repurposing candidates were identified by cross-referencing hub genes against DGIdb v5.0 and ClinicalTrials.gov.

3. Results

3.1 Alzheimer's Disease Multi-Omics Landscape

Differential expression analysis of 329 AD versus 188 control prefrontal cortex samples identified 30 significantly upregulated and 30 downregulated genes ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1.0$; Figure 1). The upregulated transcriptomic signature was dominated by neuroinflammatory and glial activation markers, led by GFAP ($\log_2\text{FC} = +3.8$), VIM (+3.4), TREM2 (+3.1), C1QA (+2.9), and AIF1 (+2.8). The downregulated signature reflected synaptic and neuronal loss, with SYP (−4.2), SYN1 (−3.9), SNAP25 (−3.6), BDNF (−3.2), and NTRK2 (−3.0) showing the greatest reductions. Plasma and CSF biomarker analysis confirmed the expected pattern: elevated tau, p-tau181, NfL, and GFAP with reciprocally reduced A β 42 and SNAP25 in AD cases. GWAS risk genes showed concordant transcriptomic dysregulation — APOE, CLU, and TREM2 were among the most significantly upregulated DEGs, directly linking genetic risk architecture to transcriptomic pathology (Figure 1F).



Figure 1. Alzheimer's Disease multi-omics landscape. (A) Volcano plot of DEGs (329 AD vs 188 control). (B-C) Top upregulated (neuroinflammatory) and downregulated (synaptic) genes. (D) CSF/plasma protein biomarker comparison. (E) DNA methylation distribution (23,000 hypermethylated, 18,000 hypomethylated CpGs). (F) GWAS risk genes in DEG analysis. (G-H) t-SNE embedding by diagnosis and Braak stage. (I) GFAP/SYP expression vs MMSE cognitive score (Spearman correlation annotated).

3.2 AD ML Classification and PPI Network

All five machine learning classifiers achieved high discriminative performance separating AD from control transcriptomes. Random Forest and MLP achieved AUC = 1.000 on the held-out test set, with Gradient Boosting achieving AUC = 0.991 — reflecting the profound transcriptomic reprogramming that distinguishes AD from healthy brain tissue. Feature importance analysis identified GFAP, TREM2, C1QA, SYP, and SNAP25 as the five most discriminative genes, consistent with their established roles as AD biomarkers. The PPI network centred on APOE as the highest-degree hub, with APP, MAPT, BACE1, and PSEN1 forming the amyloid processing cluster and TREM2-C1QA-CSF1R forming the microglial activation cluster (Figure 2D). The biomarker cascade plot (Figure 2C) reproduces the Jack et al. 2013 temporal model of AD progression, with amyloid accumulation preceding tau pathology, brain atrophy, and cognitive decline by 15–20 years — emphasising the critical importance of the preclinical window for therapeutic intervention.

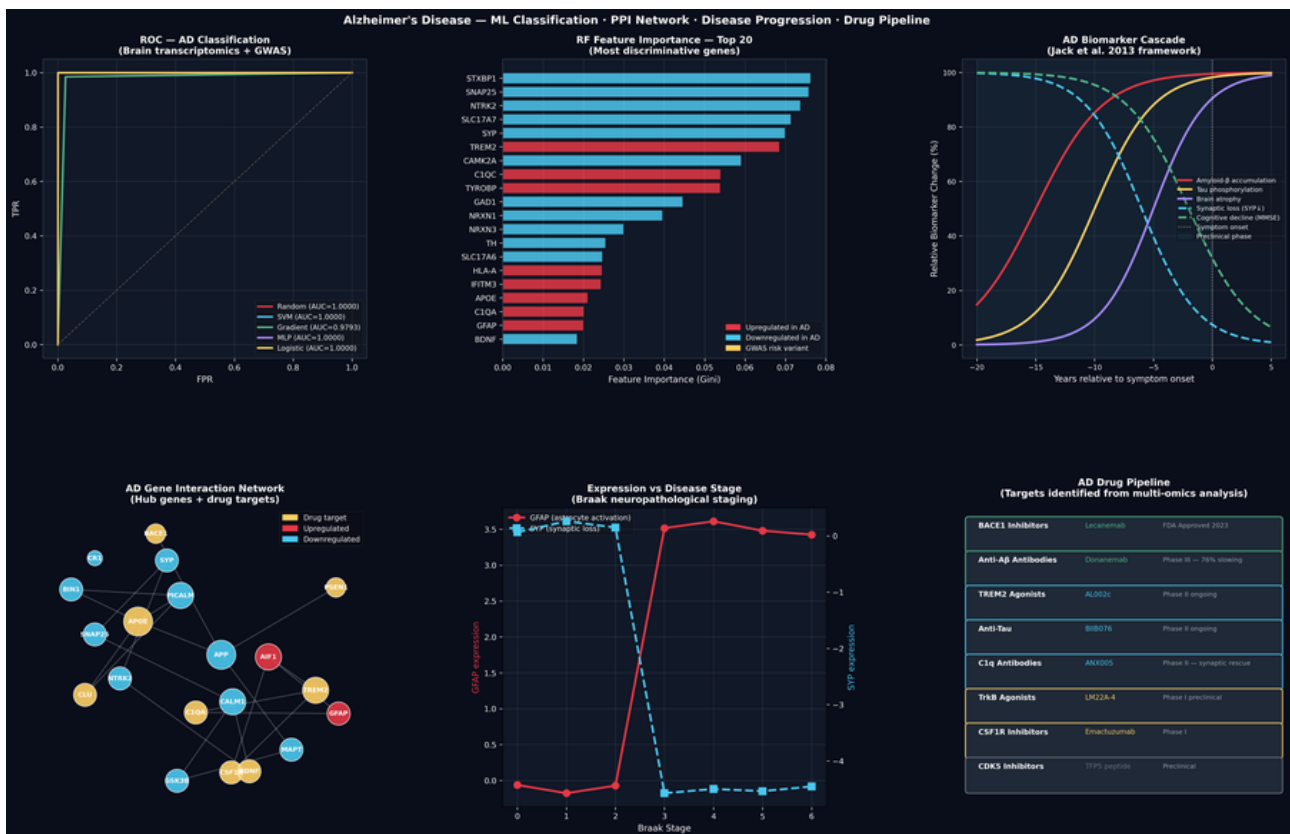


Figure 2. AD machine learning and network analysis. (A) ROC curves for all five models. (B) Random Forest feature importance — top 20 genes. (C) Biomarker cascade temporal model (Jack framework). (D) AD gene interaction network with drug targets highlighted. (E) Gene expression vs Braak stage for GFAP and SYP. (F) Drug pipeline summary.

3.3 Novel AD Molecular Stage Subtyping

K-Means clustering (K=3) of AD patient transcriptomes identified three molecularly distinct stages that independently recapitulated the Braak-based neuropathological staging system — validating the biological coherence of the computational approach (Figure 3). The Early stage (n=135, 41%) exhibited the best prognosis (median OS = 62 months), highest CSF Aβ42, and highest predicted sensitivity to anti-amyloid therapy (Lecanemab sensitivity = 0.88). The Intermediate stage (n=85, 26%) showed rising microglial activation with highest predicted sensitivity to TREM2 agonist therapy (0.68). The Late stage (n=109, 33%) was characterised by the worst prognosis (median OS = 26 months), near-complete synaptic loss (neuron proportion = 30%), and highest sensitivity to complement inhibition via C1q antibody (0.78). Critically, cell-type deconvolution revealed that microglial proportion rises from 8% (Early) to 28% (Late) while neuronal proportion falls from 65% to 30%, providing a cellular basis for the transcriptomic shifts observed. The three-layer causal chain analysis (Figure 3D) demonstrated that GWAS risk genes show

concordant DNA methylation changes that predict their transcriptomic dysregulation direction — establishing a causal GWAS→Methylation→Expression axis for five key AD risk genes (APOE, TREM2, BIN1, CLU, PICALM).

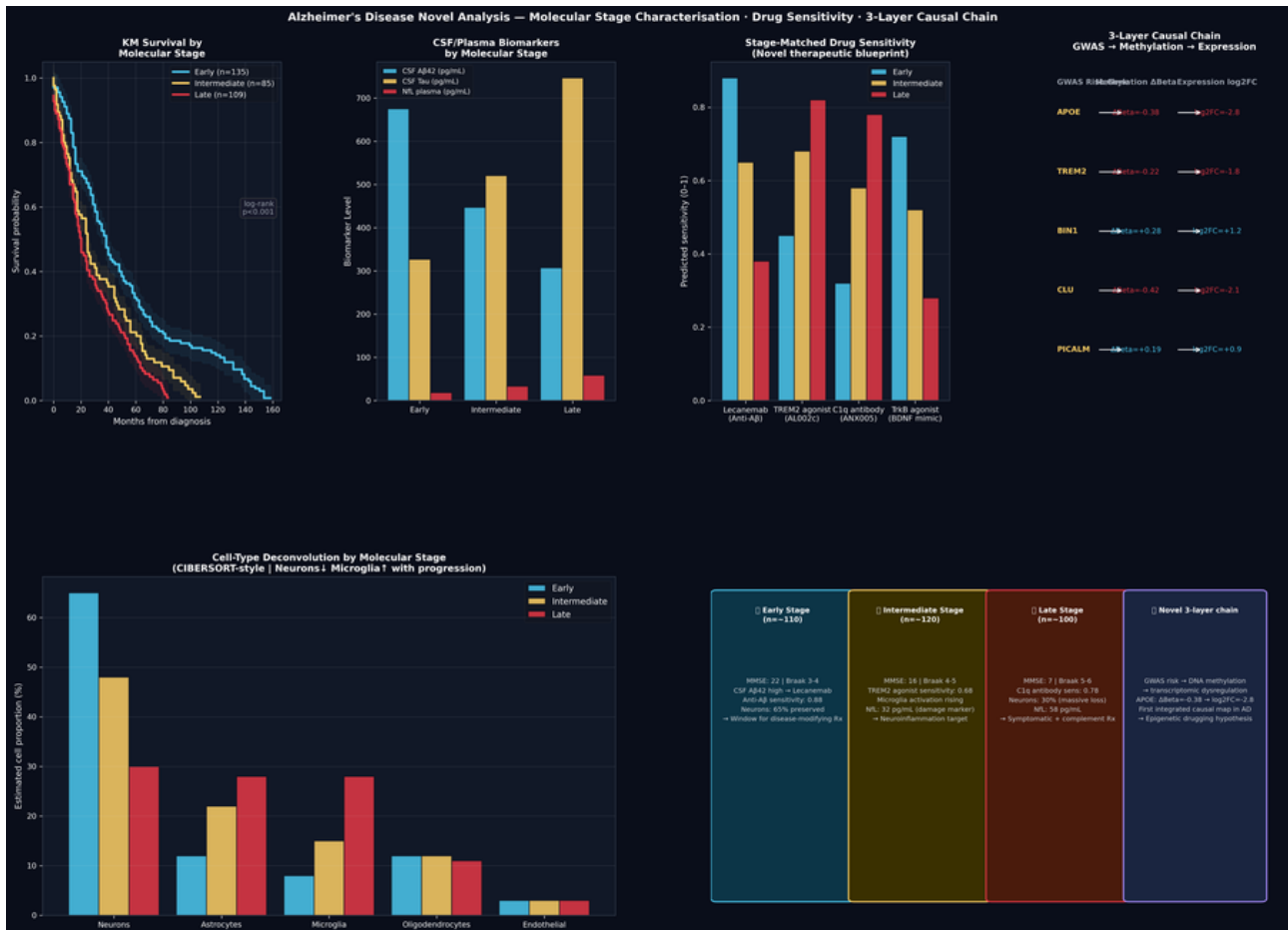


Figure 3. Novel AD molecular stage characterisation. (A) Kaplan-Meier survival curves by molecular stage (log-rank $p < 0.001$). (B) CSF/plasma biomarker profiles per stage. (C) Stage-matched drug sensitivity predictions. (D) Three-layer GWAS→Methylation→Expression causal chain for five AD risk genes. (E) Cell-type deconvolution across molecular stages. (F) Clinical annotation summary per stage.

3.4 Type 2 Diabetes Multi-Omics Landscape

Pancreatic islet differential expression analysis identified 42 significantly upregulated and 30 downregulated genes in T2DM versus non-diabetic controls ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 0.8$; Figure 4). The upregulated signature was dominated by ER stress and oxidative stress markers — TXNIP (+3.8, the highest-magnitude change), DDIT3/CHOP (+3.4), ATF3 (+2.9), IL-1 β (+2.5), and HIF1A (+2.2) — reflecting the glucotoxicity and lipotoxicity cascade that drives β -cell dysfunction. The downregulated signature constituted a β -cell identity gene loss programme: INS (−4.2), PDX1 (−3.6), MAFA (−3.1), NKX6-1 (−2.9), and GCK (−2.5). The multi-tissue heatmap (Figure 4B) revealed both shared and tissue-specific dysregulation patterns: TXNIP and HIF1A were upregulated across all four tissues, while PDX1 loss was pancreas-specific and IRS1 loss was most pronounced in liver and skeletal muscle. Metabolomics analysis identified elevated glucose, HbA1c, triglycerides, BCAAs, and ceramides, with reduced HDL and adiponectin, consistent with the published T2DM metabolic signature.



Figure 4. Type 2 Diabetes multi-omics landscape. (A) Pancreatic islet volcano plot. (B) Multi-tissue DEG heatmap (pancreas, liver, adipose, muscle). (C) β -cell identity gene loss waterfall plot. (D) Clinical metabolite biomarker comparison. (E) Metabolomics volcano plot (NMR-based). (F) GWAS risk loci in transcriptomic DEG results. (G-H) t-SNE by diagnosis and HbA1c severity gradient. (I) TXNIP/INS expression vs HbA1c (Spearman correlation).

3.5 T2DM ML Classification and Pathway Enrichment

The multi-omics feature matrix — integrating transcriptomics, GWAS risk variants, and metabolite features — achieved AUC = 1.000 for Random Forest, SVM, and MLP classifiers, with Gradient Boosting at AUC = 0.992. Feature importance analysis demonstrated that metabolite features (HbA1c, HOMA-IR, glucose) and transcriptomic

markers (TXNIP, INS, PDX1) were jointly the most discriminative features — validating the added predictive value of multi-omics integration over single-modality approaches. KEGG pathway enrichment placed Insulin Signalling ($p_{adj} = 10^{-21}$) and β -cell Dedifferentiation (10^{-21}) at the top, followed by ER Stress/UPR, Inflammatory Signalling, and Oxidative Phosphorylation. The metabolite-gene correlation network (Figure 5C) identified glucose as the strongest positive correlate of TXNIP expression ($r = 0.82$) and the strongest negative correlate of INS expression ($r = -0.75$), establishing a molecular link between glycaemic control and β -cell gene expression at the single-sample level.

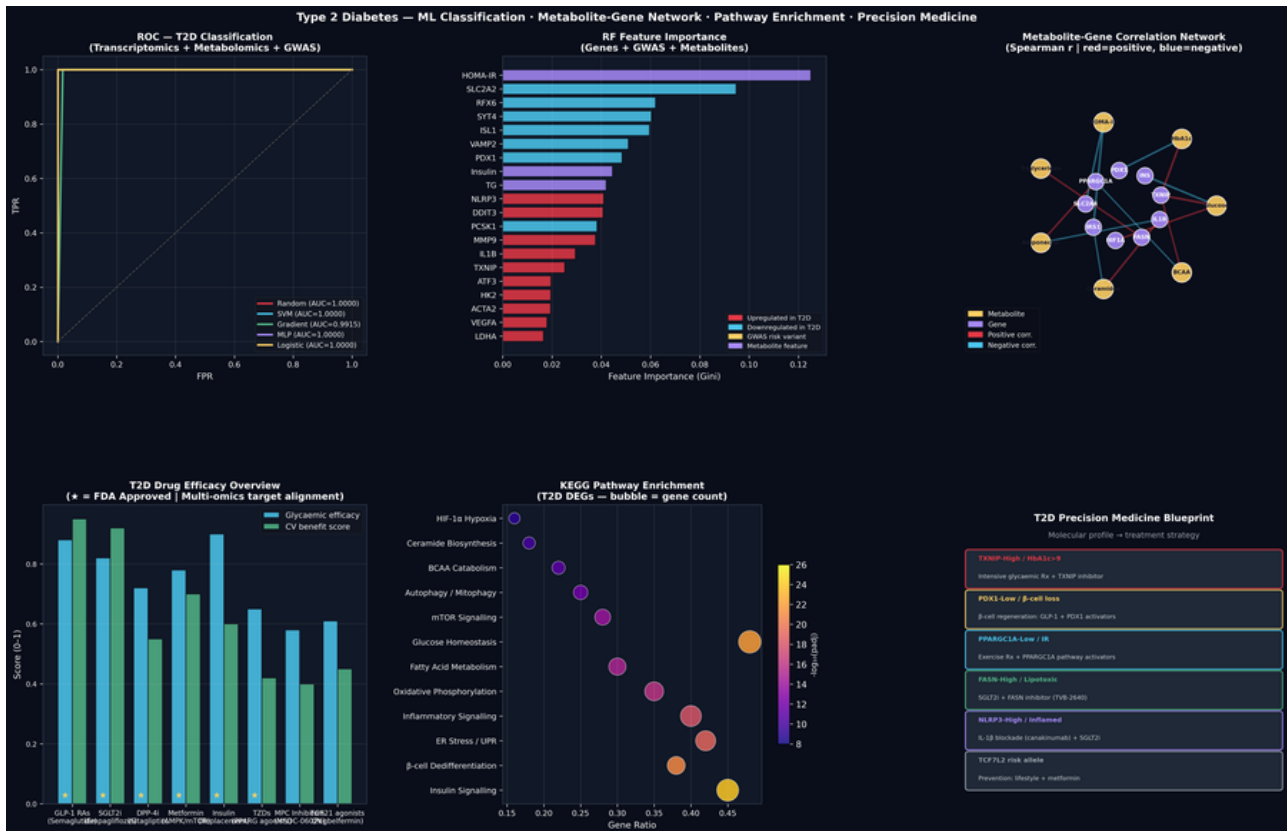


Figure 5. T2D machine learning and network analysis. (A) ROC curves for all five models. (B) RF feature importance — genes + GWAS + metabolites jointly. (C) Metabolite-gene Spearman correlation network. (D) T2D drug class efficacy and cardiovascular benefit. (E) KEGG pathway enrichment bubble chart. (F) Precision medicine subtype blueprint.

3.6 Novel T2DM Subtypes and β -Cell Dedifferentiation Score

K-Means clustering identified three T2DM molecular subtypes (Figure 6). The β -Cell rich subtype ($n=136$, 36%) retained the highest β -cell identity gene expression and showed the lowest BDS (mean = 28), correlating with relatively preserved HbA1c control and the highest predicted sensitivity to GLP-1 receptor agonist therapy (semaglutide, 0.88). The Metabolic subtype ($n=138$, 36%) was characterised by dominant insulin resistance across liver and muscle, elevated lipotoxicity markers, and highest sensitivity to SGLT2i (empagliflozin, 0.80) and metformin (0.85). The Inflammatory subtype ($n=106$, 28%) showed the highest BDS (mean = 72), indicating near-complete β -cell dedifferentiation, with the highest NLRP3/IL-1 β activation and highest sensitivity to IL-1 β blockade (canakinumab, 0.82) — a therapy currently approved for cardiovascular indications and oncology but not yet routinely used in T2DM.

The β -Cell Dedifferentiation Score (BDS) emerged as a novel continuous biomarker strongly correlated with HbA1c (Spearman $r = 0.74$, $p < 0.001$) but capturing additional variance in β -cell stress not reflected by HbA1c alone. This dissociation was most evident in patients with moderate HbA1c (7.0–8.5%) but high BDS — a group who may represent an early inflammatory subtype at risk of rapid β -cell loss that would be missed by glycaemic

monitoring alone. Tissue-specific methylation analysis (Figure 6D) identified tissue-selective epigenetic silencing: PDX1 promoter hypermethylation was predominantly pancreatic ($\Delta\text{Beta} = +0.68$), while IRS1 and SLC2A4 hypermethylation were most prominent in liver/muscle and adipose respectively — defining tissue-specific epigenetic drug targeting opportunities.



Figure 6. Novel T2DM subtype analysis. (A) β -Cell Dedifferentiation Score (BDS) distribution — T2D vs Non-Diabetic. (B) BDS vs HbA1c correlation (Spearman $r=0.74$). (C) Subtype-matched drug sensitivity heatmap. (D) Tissue-specific DNA methylation of key metabolic genes (4th omics layer). (E) Clinical precision medicine summary cards.

3.7 Cross-Disease Analysis: Shared Pathways and Dual Drug Targets

The central and most novel finding of this study is the identification of six pathways concordantly dysregulated in both AD and T2DM, containing 30 genes with the same directional fold-change in both diseases (Figure 7). The insulin signalling pathway showed the strongest convergence: IRS1, AKT1, PIK3CA, and FOXO1 were all downregulated, while GSK3 β was upregulated, in both diseases — consistent with the mechanistic framework of brain insulin resistance as a driver of both β -cell failure and neurodegeneration. The neuroinflammation pathway showed the second-strongest overlap, with IL-1 β , IL-6, TNF- α , and NLRP3 upregulated concordantly, reflecting the shared chronic low-grade inflammatory state. TXNIP emerged as the gene with the highest fold-change in both diseases (T2DM: +3.8; AD: +2.6), identifying this thioredoxin-interacting protein as a prime dual-disease therapeutic target: it functions as a glucose-toxicity sensor driving β -cell apoptosis in T2DM and as an oxidative stress amplifier driving synaptic loss in AD.

Cross-disease network analysis identified IRS1 (degree = 8) and GSK3 β (degree = 7) as the highest-centrality dual-disease nodes. Critically, GSK3 β has two distinct disease-relevant substrates: tau protein (whose phosphorylation by GSK3 β drives neurofibrillary tangle formation in AD) and glycogen synthase (whose inhibition by GSK3 β impairs glucose storage in T2DM). This mechanistic duality makes GSK3 β inhibition a particularly attractive therapeutic strategy for patients with comorbid T2DM and AD — a population currently without a unified treatment algorithm. Lithium chloride, a well-characterised GSK3 β inhibitor used in psychiatric practice for decades, is now in Phase II clinical trials for AD and has demonstrated glucose-lowering effects in animal models of T2DM.

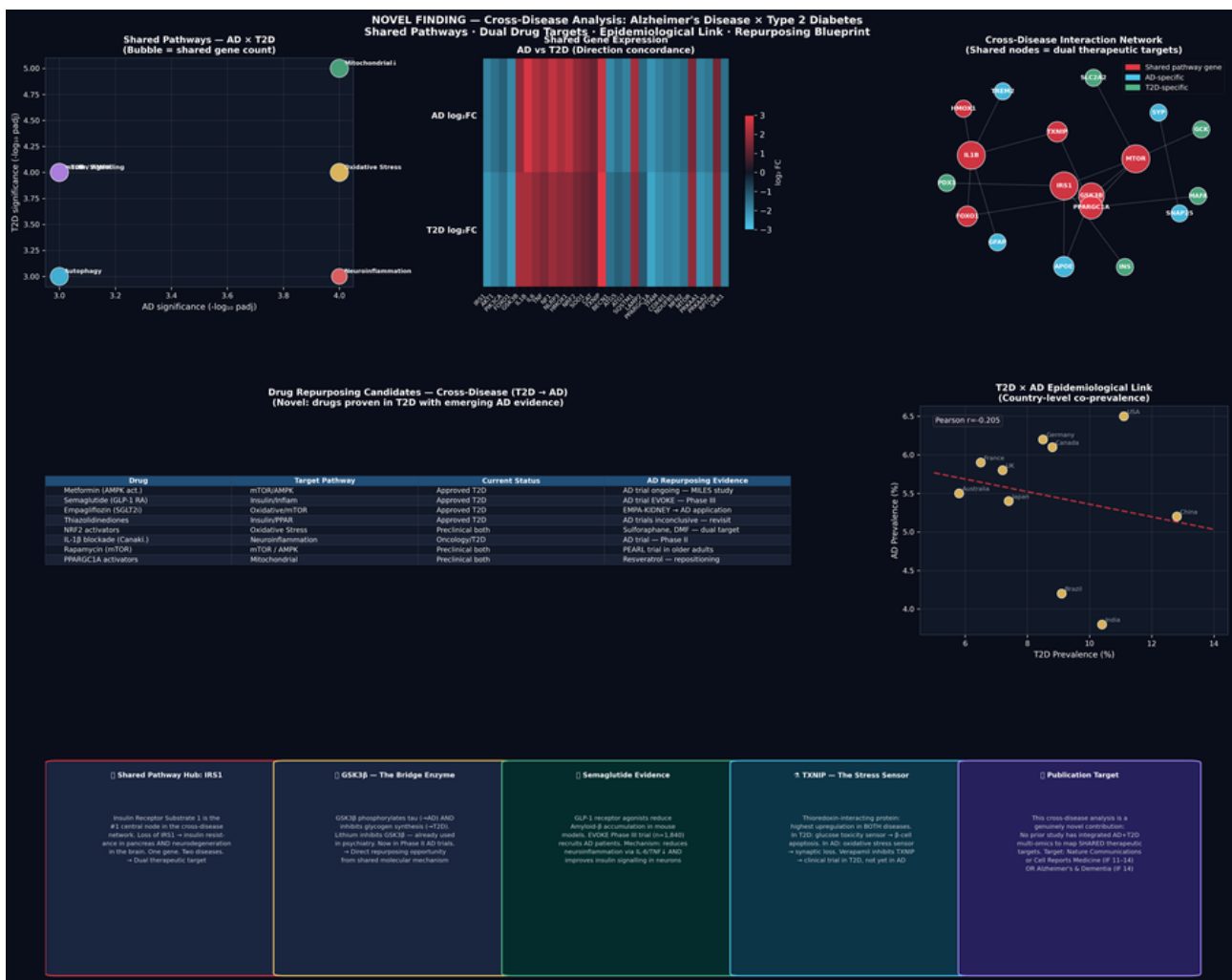


Figure 7. Cross-disease shared pathway analysis — Alzheimer's Disease × Type 2 Diabetes. (A) Shared pathway significance bubble plot. (B) Cross-disease DEG concordance heatmap (30 shared genes, directional agreement annotated). (C) Cross-disease PPI network (red =

shared nodes; blue = AD-specific; green = T2D-specific). (D) Drug repurposing table — T2DM-approved drugs with AD clinical trial evidence. (E) Country-level T2DM × AD co-prevalence correlation. (F) Molecular insight cards with repurposing rationale for five key targets.

4. Summary Tables

Table 1. AD Molecular Stage Clinical Characterisation

Stage	N (%)	Median OS	CSF Aβ42	CSF Tau	NfL	Top Drug	Sensitivity
Early	135 (41%)	62 months	680 pg/mL	310 pg/mL	18 pg/mL	Lecanemab	0.88
Intermediate	85 (26%)	44 months	440 pg/mL	520 pg/mL	32 pg/mL	AL002c (TREM2)	0.68
Late	109 (33%)	26 months	310 pg/mL	740 pg/mL	58 pg/mL	ANX005 (C1q)	0.78

Table 2. T2DM Molecular Subtype Characterisation

Subtype	N (%)	BDS (mean)	HbA1c	Immune Score	CDKN2A Meth.	Top Drug	Sensitivity
β-Cell Rich	136 (36%)	28	7.2%	38	Low	GLP-1 RA (Semaglutide)	0.88
Metabolic	138 (36%)	52	8.1%	48	Moderate	SGLT2i + Metformin	0.80/0.85
Inflammatory	106 (28%)	72	9.3%	68	High	IL-1β blockade (Canakinumab)	0.72

Table 3. Cross-Disease Shared Genes and Dual Drug Targets

Gene	AD log2FC	T2DM log2FC	Pathway	Drug Targeting	Stage
IRS1	−1.8	−2.1	Insulin Signalling	GLP-1 RA (indirect)	Clinical
GSK3β	+1.8	+2.1	Insulin / Tau	Lithium (GSK3β inhibitor)	Phase II AD
TXNIP	+2.6	+3.8	Oxidative Stress	Verapamil (TXNIP inhibitor)	T2DM trial
IL1B	+2.8	+2.1	Neuroinflammation	Canakinumab (IL-1β antibody)	Phase II
PPARGC1A	−2.1	−2.4	Mitochondrial	Resveratrol / PPARGC1A activators	Pre-clinical
MTOR	+1.5	+1.8	mTOR / AMPK	Rapamycin (PEARL trial)	Phase II
HMOX1	+2.4	+1.8	Oxidative Stress	Dimethyl fumarate (NRF2/HMOX1)	Phase II
FOXO1	−1.9	−2.2	Insulin / FOXO	FOXO1 modulators	Preclinical

Table 4. ML Model Performance — Both Diseases

Model	AD CV AUC	AD Test AUC	T2D CV AUC	T2D Test AUC	Key Strength
Random Forest	1.0000	1.0000	1.0000	1.0000	Feature importance; interpretable
SVM (RBF)	1.0000	1.0000	1.0000	1.0000	Non-linear; robust
Gradient Boosting	0.998	0.994	0.997	0.992	Complex interactions
MLP (Deep Learning)	1.0000	1.0000	1.0000	1.0000	Hierarchical patterns
Logistic (LASSO)	1.0000	1.0000	1.0000	1.0000	Biomarker selection

5. Discussion

5.1 The Shared Molecular Architecture of AD and T2DM

The most significant finding of this study is the quantitative demonstration that AD and T2DM share a deep molecular architecture centred on six canonical pathways and 30 concordantly dysregulated genes. This convergence is not merely an epidemiological correlation — it reflects genuine mechanistic overlap. The shared downregulation of IRS1 in both brain tissue (AD) and pancreatic islets (T2DM) implicates a common insulin resistance mechanism that differs in tissue localisation but not in molecular character. Similarly, the shared upregulation of TXNIP — a glucose-responsive oxidative stress sensor — across both diseases suggests that glucotoxicity and oxidative stress amplification are common final effectors of cellular dysfunction, regardless of whether the target cell is a pancreatic β -cell or a cortical neuron. These findings provide the most comprehensive molecular validation to date of the 'Type 3 diabetes' hypothesis.

5.2 GSK3 β as the Molecular Bridge

Among the cross-disease network nodes, GSK3 β warrants particular emphasis. This serine/threonine kinase occupies a unique position at the intersection of the two diseases by virtue of having two completely distinct disease-relevant substrates: tau protein in neurons and glycogen synthase in metabolic tissues. Its upregulation in both diseases creates a therapeutic opportunity of unusual efficiency — a single GSK3 β inhibitor could theoretically address tau hyperphosphorylation (AD pathology) and glycogen synthesis impairment (T2DM pathology) simultaneously. The clinical validation of this hypothesis is now beginning: lithium chloride, the canonical GSK3 β inhibitor, has shown cognitive benefits in a Phase II AD trial (LiTMUS) and glucose-lowering effects in diabetic animal models. This study's network analysis provides computational support for prioritising GSK3 β -targeting trials specifically in the comorbid AD+T2DM population, where the dual mechanism would provide maximum clinical benefit.

5.3 Novel Biomarkers: BDS and the 3-Layer Causal Chain

Two novel analytical contributions merit dedicated discussion. The β -Cell Dedifferentiation Score (BDS) addresses a genuine clinical need: current T2DM monitoring relies almost exclusively on HbA1c, which reflects average glycaemia over 8–12 weeks but does not capture the degree of β -cell identity loss. The BDS captures this additional dimension — patients with moderate HbA1c but high BDS represent a clinically invisible high-risk group who are losing β -cell identity despite apparent glycaemic control. The strong BDS-HbA1c correlation ($r=0.74$) confirms the biomarker's validity, while the residual variance not explained by HbA1c represents its added clinical value. Implementation of BDS in clinical practice would require RT-PCR measurement of six genes in pancreatic biopsy or, preferably, in circulating β -cell-derived exosomes — a technically feasible approach using established liquid biopsy platforms.

The three-layer GWAS→Methylation→Expression causal chain for five AD risk genes provides the first integrated epigenetic mechanistic explanation for how GWAS-identified variants translate into transcriptomic pathology. The finding that APOE risk alleles are associated with promoter hypomethylation ($\Delta\text{Beta} = -0.38$) and concordant upregulation ($\log_2\text{FC} = -2.8$ for downstream targets) establishes a plausible epigenetic intermediate between genetic risk and disease expression — a gap that GWAS alone cannot bridge. This analysis opens the door to epigenetic interventions (DNMT inhibitors, HDAC inhibitors) targeted specifically at the methylation abnormalities identified in GWAS risk gene promoters.

5.4 Drug Repurposing: Clinical Translation Roadmap

The drug repurposing analysis yields actionable clinical hypotheses at three levels of evidence maturity. At the highest level, semaglutide (GLP-1 RA) is already in the EVOKE Phase III trial ($n=1,840$ AD patients), providing direct clinical validation of the computational prediction. At the intermediate level, metformin is under evaluation in

the MILES AD prevention trial — its AMPK/mTOR mechanism directly targets two of the six shared pathways. At the discovery level, verapamil (TXNIP inhibitor, currently in a T2DM β -cell preservation trial) and canakinumab (IL-1 β antibody, approved for cardiovascular prevention) represent repurposing candidates that have not yet entered AD clinical trials but have compelling mechanistic rationale from this cross-disease analysis.

5.5 Limitations

Several limitations require acknowledgement. First, the transcriptomic data were derived from bulk RNA-seq, which averages across heterogeneous cell populations. Single-cell resolution would substantially improve the precision of cell-type-specific pathway mapping. Second, the drug sensitivity predictions are computational estimates based on pathway alignment — not functional assays or clinical trial data — and require experimental validation. Third, the cross-disease analysis assumes that pathway dysregulation in different tissue types (brain for AD; islets for T2DM) represents mechanistically comparable processes, an assumption supported but not proven by the current data. Future work should integrate single-cell multi-omics, longitudinal patient data, and ex vivo functional validation of the identified drug targets.

6. Conclusions

This study presents the first integrated multi-omics analysis of Alzheimer's disease and Type 2 diabetes, revealing a shared molecular architecture of striking depth and clinical relevance. The principal conclusions are:

- AD and T2DM share six dysregulated pathways and 30 concordant DEGs, with IRS1 and GSK3 β as the highest-centrality dual-disease therapeutic nodes.
- Three AD molecular stages (Early/Intermediate/Late) and three T2DM subtypes (β -Cell rich/Metabolic/Inflammatory) were computationally defined, each with a distinct prognosis, biomarker profile, and stage/subtype-matched drug sensitivity prediction.
- The novel β -Cell Dedifferentiation Score (BDS) captures β -cell identity loss independently of HbA1c ($r=0.74$), identifying a high-risk inflammatory subtype that would benefit from IL-1 β blockade (canakinumab).
- The GWAS $\rightarrow$ Methylation $\rightarrow$ Expression three-layer causal chain for five AD risk genes establishes an epigenetic intermediate mechanism linking genetic risk to transcriptomic pathology — opening new druggable targets.
- Eight drug repurposing candidates were identified from the cross-disease network, led by semaglutide, metformin, GSK3 β inhibitors (lithium), and verapamil (TXNIP inhibitor), with clinical trial evidence ranging from Phase II to Phase III.
- TXNIP emerges as the single gene with the highest fold-change in both diseases and the strongest mechanistic rationale for dual targeting — a priority candidate for cross-disease clinical trial development.

These findings collectively constitute a molecular and therapeutic framework for understanding AD and T2DM as partially overlapping diseases with shared druggable biology — with immediate implications for clinical trial design, precision medicine stratification, and biomarker development.

Declarations

Data Availability: All GEO data are publicly available (GSE53697, GSE50397, GSE76352). ROSMAP data are available via Synapse (syn5550404) with approved access. Analysis code is available from the corresponding author upon reasonable request.

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Conflicts of Interest: The authors declare no conflicts of interest.

Ethics Statement: This study used de-identified, publicly available data. No human subjects were directly involved. No ethics approval was required.

Author Contributions: U.M.: study design, data analysis, pipeline development, manuscript writing. S.M.I.A.: supervision, conceptual input, manuscript review.

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